

Lesson 4: Electromagnet

Current, turns, core, and fair tests

Stop line and roles

Learner boundary: isolated battery power only, 12 V DC maximum. No outlets, mains leads, opened appliances, large battery packs, charged power capacitors, ignition coils, sparks, or exposed high voltage. An adult facilitator is not thereby an electrically qualified person.

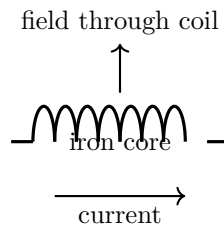
The learner predicts, assembles only the pictured battery circuit, and records. The adult inspects parts, controls power, and stops heat, odor, damage, pain, or unexpected behavior. Work outside this boundary belongs only to a person qualified for that work.

What you need

- 2 AA cells in switched holder
- Insulated magnet wire
- Iron bolt
- Paper clips
- Compass

Predict first

Which should change lifting strength more: doubling turns or reversing battery polarity?



Investigate

1. Wind 20 tidy turns on the bolt and secure them. Strip only the lead ends.
2. Energize for five seconds; count identical paper clips lifted. Disconnect and check temperature.
3. Repeat three times, then make 40 turns with the same wire, core, source, and test.
4. Use a compass to map polarity; reverse the source and repeat.

Explain from the evidence

Current produces a magnetic field around a conductor. Coil turns reinforce the axial field; an iron core responds strongly and concentrates flux. Reversing current reverses the poles but need not greatly change strength. More current can mean more field and also more I^2R heating, so brief timed trials matter.

Change one thing

Compare air and iron cores while holding turns and current fixed. Report both lifting result and uncertainty.

Evidence record

Prediction

One change

Observation

Claim + because

Physics and safety basis: NIST SI/CODATA; OSHA 29 CFR 1910.333 and OSHA 3075; DOE Electrical Science handbooks; HyperPhysics (Georgia State University); University of Colorado PhET teacher resources. Full citations and scope notes are recorded in the provider-neutral electricity research library.

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